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BATCH:-A1
ASSINGMENT:-3

Aim:-Data Wrangling with pandas: Manipulate and transform data to suit the analysis needs.

Theory:- Q.1.What is data wrangling? Explain the difference between data wrangling and data preprocessing. **Answer:-**Data wrangling (also called data munging) is the process of cleaning, transforming, and organizing raw data into a structured and usable format for analysis. It involves tasks like handling missing values, correcting errors, filtering data, merging datasets, and converting data types so that the data becomes accurate and consistent.

Difference between Data Wrangling and Data Preprocessing: Data Wrangling: It focuses on cleaning and preparing raw, messy data from different sources. The main goal is to make data usable. It includes tasks like removing duplicates, fixing inconsistencies, and reshaping data. Data Preprocessing: It is a broader step used mainly before applying machine learning models. It includes transforming already cleaned data into a format suitable for algorithms. This involves normalization, scaling, encoding categorical variables, and feature selection.

Q.2.What is the importance of exploratory data analysis (EDA) in Pandas? **Answer:-**Exploratory Data Analysis (EDA) in Pandas is an important step used to **understand, summarize, and analyze a dataset before applying any statistical or machine learning techniques**. **bold text** It helps in gaining insights into the structure, patterns, and relationships within the data. Using Pandas functions like `head()`, `info()`, `describe()`, and `value_counts()`, analysts can quickly examine data types, detect missing values, and understand distributions.

The importance of EDA lies in identifying **errors, outliers, and inconsistencies** in the dataset, which improves data quality. It also helps in discovering **trends, correlations, and patterns**, which guide decision-making and model selection. Additionally, EDA ensures that the data is suitable for further processing and reduces the chances of incorrect conclusions. Overall, EDA in Pandas makes data analysis more accurate, efficient, and meaningful by providing a clear understanding of the dataset before deeper analysis.

Q.3.What are the main differences between Series, DataFrame, and Panel? Answer:- The main differences between Series, DataFrame, and Panel in Pandas are based on their structure and dimensionality.

Series: A Series is a one-dimensional (1D) labeled array that can hold data of any type (integers, floats, strings, etc.). It is similar to a single column in a table, where each value has an associated index. DataFrame: A DataFrame is a two-dimensional (2D) labeled data structure with rows and columns, similar to a table or spreadsheet. It can store different data types in different columns and is the most commonly used Pandas structure for data analysis. Panel: A Panel is a three-dimensional (3D) data structure, which was used to store multiple DataFrames. However, it is now deprecated (removed) in modern versions of Pandas, and users are encouraged to use alternatives like multi-index DataFrames or libraries such as xarray.

Q.4.Explain the use of: a. isnull() b. notnull() c. dropna() d. fillna() **Answer:-**In Pandas, handling missing data is very important. The following functions help detect and manage missing (NaN) values:

a. isnull() This function is used to detect missing values in a dataset. It returns True for missing (NaN) values and False for non-missing values. Useful for identifying where data is missing.

b. notnull() This is the opposite of isnull(). It returns True for non-missing values and False for missing ones. Helps in filtering valid data.

c. dropna() This function is used to remove missing values from a dataset. It can drop rows or columns that contain NaN values. Useful when you want to clean the dataset by deleting incomplete data.

d. fillna() This function is used to replace missing values with a specified value (like 0, mean, median, etc.). Useful when you want to keep the data but handle missing entries properly.

** Q.5.What is column transformation and why is it important in data analysis?** **Answer:-**Column transformation is the process of modifying or converting the values of a column in a dataset into a different format or scale to make it more suitable for analysis. This can include operations like normalization, scaling, encoding categorical data, applying mathematical functions (log, square root), or creating new derived columns from existing ones.

The importance of column transformation in data analysis lies in improving the quality, consistency, and usability of data. It helps in handling skewed data distributions, bringing different variables to a common scale, and converting non-numeric data into numeric form required for machine learning models. Proper transformation also enhances model performance, makes patterns more visible, and ensures accurate results.

```
#CODE

#Add a new column with constant values.
import pandas as pd
```

```
df = pd.DataFrame({
    'Name': ['A', 'B', 'C'],
    'Marks': [80, 85, 90]
})

df['Grade'] = 'Pass'

print(df)
```

```
  Name  Marks Grade
0    A     80  Pass
1    B     85  Pass
2    C     90  Pass
```

```
#Group data by a categorical column and calculate mean.
import pandas as pd

df = pd.DataFrame({
    'Department': ['IT', 'HR', 'IT', 'HR', 'Finance'],
    'Salary': [50000, 40000, 60000, 45000, 55000]
})

result = df.groupby('Department')['Salary'].mean()

print(result)
```

```
Department
Finance    55000.0
HR         42500.0
IT         55000.0
Name: Salary, dtype: float64
```

```
#Merge two DataFrames on a common column.
import pandas as pd

df1 = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['A', 'B', 'C']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 3],
    'Marks': [80, 85, 90]
})

merged_df = pd.merge(df1, df2, on='ID')

print(merged_df)
```

```
   ID Name  Marks
0   1    A     80
1   2    B     85
2   3    C     90
```

```
#Perform pivot and melt operations on a dataset.
import pandas as pd

df = pd.DataFrame({
    'Name': ['A', 'A', 'B', 'B'],
    'Subject': ['Math', 'Science', 'Math', 'Science'],
    'Marks': [80, 85, 90, 95]
})

pivot_df = df.pivot(index='Name', columns='Subject', values='Marks')

print(pivot_df)
```

```
Subject  Math  Science
Name
A          80       85
B          90       95
```

```
#Perform time series based data wrangling.
import pandas as pd

data = {
    'Date': ['2024-01-01', '2024-01-02', '2024-01-03', '2024-01-04'],
    'Sales': [200, 250, 300, 280]
}

df = pd.DataFrame(data)
```

```
df['Date'] = pd.to_datetime(df['Date'])

df.set_index('Date', inplace=True)

print("Original Data:\n", df)

df['Month'] = df.index.month
df['Day'] = df.index.day

filtered = df[df.index > '2024-01-02']

resampled = df['Sales'].resample('2D').mean()

print("\nFiltered Data:\n", filtered)
print("\nResampled Data:\n", resampled)
```

```
Original Data:
             Sales
Date
2024-01-01    200
2024-01-02    250
2024-01-03    300
2024-01-04    280

Filtered Data:
             Sales  Month  Day
Date
2024-01-03    300      1    3
2024-01-04    280      1    4

Resampled Data:
 Date
2024-01-01    225.0
2024-01-03    290.0
Freq: 2D, Name: Sales, dtype: float64
```

Conclusion:-In conclusion, data wrangling with Pandas is a crucial step in the data analysis process, as it involves cleaning, transforming, and organizing raw data into a structured and usable format. By using Pandas functions, analysts can efficiently handle missing values, reshape datasets, filter data, and perform transformations that improve data quality and consistency. This ensures that the data is accurate, reliable, and ready for further analysis or modeling. Ultimately, effective data wrangling enhances the quality of insights and supports better decision-making.